OpenBuild — Executive Findings

Analytics engineering portfolio · 108K orders · 87K users · 2019–2022 · Medallion architecture · 37 tested transformations

1. Loyalty's retention deficit is a channel problem, not a program problem

RESULT

1a. Loyalty membership at acquisition correlates with 3.6× *worse* retention. Loyalty members retain at 0.45% in month 1 vs. 1.63% for non-loyalty (averaged across 48 cohorts). Deficit is front-loaded; converges by month 9. Mechanism: loyalty disproportionately recruits Airpods buyers (58.1% of loyalty first-purchases vs. 36.7% non-loyalty) and underweights replenishables (5.7% Charging Cable Pack vs. 33.6% non-loyalty). AOVs roughly equal ($240 vs. $259) — *not* a discount effect.

1b. The email channel is the loyalty signup pipeline — and a low-quality one. Email acquires 60% loyalty members (vs. direct's 42%, affiliate's 18%) AND email-acquired loyalty members carry the lowest first-purchase AOV at $200 (vs. direct loyalty's $253). Email's non-loyalty AOV is also lowest at $142. The loyalty deficit is not caused by the program itself — it is caused by where loyalty signups originate.

IMPLICATION

The loyalty program amplifies the email channel's structural retention problem. Lever is acquisition channel mix, not program redesign. Reroute loyalty signup mechanics out of email-driven flows toward direct or affiliate channels.

CAUSES

Cause	Weight	Status
Durable goods have low repurchase frequency (1.23 orders/user across 4 years)	High	Tested
Catalog skews to one-time purchases (only 1 of 8 products is replenishable)	High	Tested
Loyalty program scaled 11.6% (2019) → 55.8% (2022), recruiting into one-and-done categories	High	Tested
Email channel concentrates loyalty signups (60%) and recruits lowest-AOV buyers ($142–$200)	High	Tested

Source models: mart_cohort_retention, mart_loyalty_retention, mart_marketing_acquisition · See Appendix A1, A1b

2. Laptops drive 2.5–4.3× the company refund rate

RESULT

Refund baseline: 4.97%. Top offenders all laptops: ThinkPad × CA (21.3%), ThinkPad × GB (16.8%), MacBook Air × CA (16.3%). MacBook Air × US: largest dollar leak, $365K refunded.

IMPLICATION

Target laptops first. Geographic variation suggests fulfillment or returns-policy investigation. Negative result: fulfillment-SLA correlation is weak (4.7% at ≤5d vs. 5.0% at 6-10d) — spec-mismatch hypothesis strengthens; fulfillment hypothesis weakens.

CAUSES

Cause	Weight	Status
High-AOV products carry higher absolute return risk	High	Tested
Spec / expectation mismatch in laptops vs. accessories	High	Partially tested
Country-level returns policy or fulfillment variance	Medium	Partially tested
Fulfillment SLA effect on refund rate	Low	Tested (negative)

Source models: mart_refund_metrics, fct_orders, dim_country · See Appendix A2

3. Mobile is structurally a low-AOV channel — a discovery surface, not a transaction one

RESULT

Website = 96.8% of lifetime revenue. Mobile generates 17% of orders but only 3% of revenue (AOV ~6.4× lower: $46 vs. $296). Mobile share moved from 2.95% (2019) to 3.93% (2022) — statistically real, business-trivially small. Even mobile-acquired users (who created their account on mobile) purchase through the website roughly half the time.

IMPLICATION

Mobile is more accurately understood as a discovery surface than a transaction surface. Investment must target AOV / conversion quality, not order volume. At current AOV, doubling mobile orders only adds ~3 pp to total revenue.

CAUSES

Cause	Weight	Status
Mobile users skew toward accessory purchases	High	Tested
Catalog product mix is desktop-shaped (laptops, monitors)	High	Tested
High-AOV purchases migrate to web checkout even from mobile-acquired users	High	Partially tested
Mobile UX friction at high-value checkout	Medium	Hypothesis

Source models: mart_channel_revenue, fct_orders, dim_platform · See Appendix A3

4. The catalog is dangerously concentrated AND wastefully fragmented

RESULT

Top 3 products = 85% of revenue: 27in 4K Gaming Monitor (35.6%), Apple Airpods (28.2%), MacBook Air (21.4%). Bottom 5 products combined = under 5% of revenue: Bose Soundsport had 27 lifetime orders ($3,339); Apple iPhone, despite a $688 AOV, accounts for 0.8% of revenue.

IMPLICATION

Two opposite strategic problems coexist. The top is dangerously concentrated — any supplier change in Apple, Samsung, or the monitor cycle erases 20–35% of revenue overnight. The bottom is wastefully fragmented — 5 products consume ops complexity while contributing negligibly. Rationalize the bottom (kill or aggressively scale Bose, iPhone) while diversifying the top.

CAUSES

Cause	Weight	Status
Marketing and merchandising over-invested in top-3 hero products	High	Tested
Long-tail products lack clear positioning or distribution channel fit	High	Tested
Apple iPhone has price-point appeal ($688 AOV) but no dedicated marketing investment	Medium	Partially tested
Bose Soundsport may be a discontinued / dead-end SKU	Medium	Hypothesis

Source models: mart_product_concentration, fct_orders, dim_product · See Appendix A4

Full analysis, SQL, and tests: github.com/joyce92200/Ecommerce-analytics-engineering

Appendix — Methodology

Each finding's metric definition, SQL logic, and caveats. All numbers reproducible from the repository above.

A1 — Cohort retention

Metric. Period-active retention: % of a cohort placing at least one order during month N after acquisition. Cohort defined by user's first purchase month: DATE_TRUNC('month', MIN(purchase_ts)).

```
retention_pct = active_users / cohort_size × 100
active_users = COUNT(DISTINCT user_id) per (cohort_month, months_since_acquisition)
cohort_size = active_users at months_since_acquisition = 0
```

Caveats. Cohorts after Dec 2021 right-censored. 3 orders (0.003%) with unparseable timestamps excluded.

A1b — Loyalty retention + email channel mechanism

Metric. Same retention definition as A1, split by loyalty status at first purchase, then crossed with marketing channel at first purchase.

Anti-confounder. Uses *loyalty_at_first_purchase* and *marketing_channel_at_first_purchase*, not current values. This avoids reverse causality where users who retained → joined loyalty later → look like good retainers. Snapshot at acquisition is the only valid measure for cohort comparison.

```
WITH first_attrs AS (
  SELECT DISTINCT user_id,
    FIRST_VALUE(loyalty_status) OVER (PARTITION BY user_id ORDER BY purchase_ts) AS first_loyalty,
    FIRST_VALUE(marketing_channel) OVER (PARTITION BY user_id ORDER BY purchase_ts) AS first_channel
  FROM stg_orders
)
SELECT first_channel, first_loyalty, COUNT(*), AVG(first_aov), ...
GROUP BY first_channel, first_loyalty
```

Sample sizes. Non-loyalty: 56,422 users acquired across 48 cohorts. Loyalty: 45,443 users. Both segments large enough for the gap to be real signal, not noise. Email channel: 18,003 users (60% loyalty share, $142–$200 AOV). Direct channel: 78,583 users (42% loyalty share, $254–$276 AOV).

Caveats. "Loyalty" is a binary 0/1 flag in source. The 2020 structural break in loyalty concentration cannot be diagnosed from this data alone. Email-channel concentration of loyalty is correlation; direction of causality (does email default into loyalty signup, or does email target loyalty-prone users?) is underdetermined.

A2 — Refund metrics + fulfillment-SLA negative result

Metric. Two metrics at (product × country) grain: *refund_rate_pct* = refunded orders / total × 100; *refunded_revenue_usd* = SUM(usd_price WHERE is_refunded = 1).

Refund definition. Order is refunded if REFUND_TS_sanity_check = 'more than 180 days'. (Updated from prior REFUND_TS-not-null logic; new flag is binary and consistent.)

```
WHERE orders >= 100
ORDER BY refund_rate_pct DESC
LIMIT 10
```

Negative result on fulfillment SLA. Tested hypothesis: longer shipping times → higher refund rates. Computed fulfillment_band = DATE_DIFF('day', purchase_ts, delivery_ts), bucketed into ≤5 / 6-10 / 11-15 / >15 days. Result: 4.7% refund at ≤5d vs. 5.0% at 6-10d — gap too small to support fulfillment-driven explanation. This redirects investment toward product-listing accuracy and pre-purchase spec clarity.

Caveats. 100-order minimum is a statistical-reliability threshold. 4 orders with NULL country_code retained as distinct segment. 18 orders with non-standard codes (EU, AP) mapped to Unclassified.

A3 — Channel revenue + device crosstab

Metric. Net revenue (gross minus refunded) by (purchase_month × purchase_platform) with within-month share %.

```
net_revenue_usd = SUM(usd_price WHERE is_refunded = 0)
share_of_month_pct = net_revenue / SUM(net_revenue) OVER (PARTITION BY purchase_month) × 100
```

Account-creation device observation. Crosstab of device_at_first_purchase × purchase_platform shows mobile-created users still buy on web ~54% of the time. This is supporting evidence inside Finding 3 rather than a standalone finding — the underlying data is the same as the channel-revenue analysis viewed at user-acquisition grain.

Caveats. Channel attribution is platform-only (website vs. mobile app). AOV comparisons use net_revenue / orders across platforms.

A4 — Product concentration (Pareto)

Metric. Per-product net revenue, share of total, and cumulative share computed via window function with explicit ROWS framing.

```
100.0 * SUM(net_revenue) OVER (
  ORDER BY net_revenue DESC
  ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
) / SUM(net_revenue) OVER () AS cumulative_pct_of_revenue
```

Why ROWS, not RANGE. The default window frame (RANGE) handles ties differently — tied revenue values would all get the same cumulative percentage, producing wrong Pareto curves. Explicit ROWS framing is the deterministic, safer choice and what most engines optimize for.

Test. Last row's cumulative_pct_of_revenue must equal 100.00 — invariant validates the window framing. (Test #36 in test_models.sql; without it, a regression in the framing clause would silently produce wrong concentration metrics.)

Caveats. Net revenue (refunds excluded) used; gross revenue would shift Bose Soundsport ranking marginally but not change concentration story.

How to verify

Clone the repository, install requirements, open notebooks/01_cohort_exploration.ipynb, and run all cells top-to-bottom. Each finding corresponds to a specific cell whose output matches the numbers reported. The 37 data quality tests in tests/test_models.sql validate every input model.